Using the SAP Java Connector in Productions

Version 2026.1
2026-04-20

InterSystems Corporation One Congress Street Boston MA 02114 www.intersystems.com

Using the SAP Java Connector in Productions
PDF generated on 2026-04-20
InterSystems IRIS Version 2026.1

For Support questions about any InterSystems products, contact:

InterSystems Worldwide Response Center (WRC)

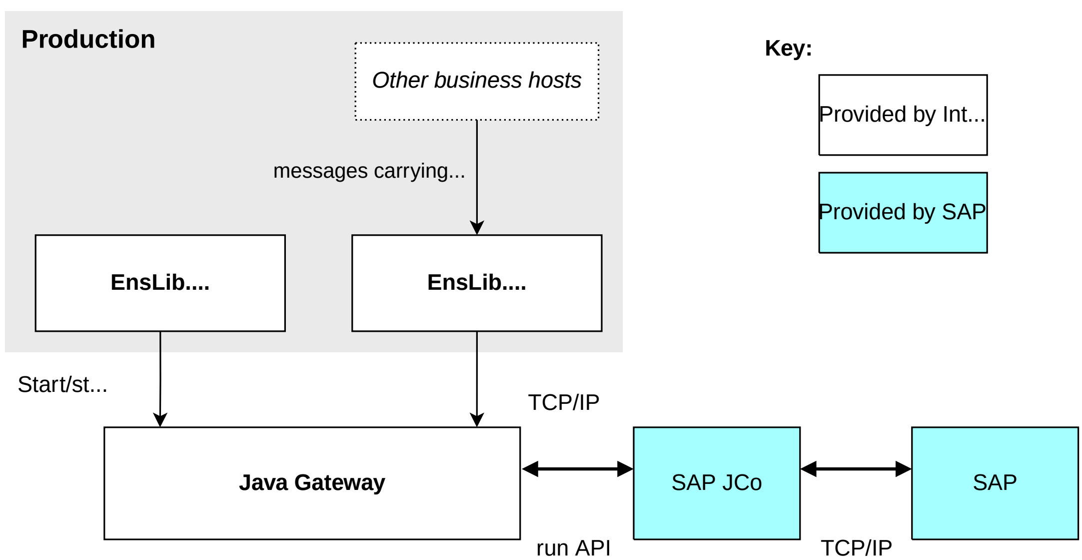
Tel:	+1-617-621-0700
Tel:	+44 (0) 844 854 2917
Email:	support@InterSystems.com

Table of Contents

1

Overview of SAP Java Connector

SAP Java Connector (SAP JCo) is a Java-based component that supports communication with an SAP Server in both directions. InterSystems provides components that you can add to a production to enable the production to communicate with SAP JCo, and thus with an SAP Server. The following picture shows the architecture:



The architecture includes the Java Gateway, which must be running.

To communicate with SAP JCo, the production must include the following items:

- EnsLib.SAP.Operation, which communicates via TCP/IP with the Java Gateway.
- EnsLib.JavaGateway.Service, which starts and stops the Java Gateway.

  This business host performs an additional function: its settings indicate the location of the Java Gateway. When correctly configured, the EnsLib.SAP.Operation business host retrieves those settings and uses them. Thus it is not necessary to set any environment variables.

  Unlike most business hosts in a production, EnsLib.JavaGateway.Service does not handle any production messages.

2
Setup Tasks for the SAP Java Connector

Before you can use the SAP components in a production, you must perform the setup activities discussed on this page.

To access SAP, it is necessary to provide a username and password. This means that you must also create production credentials that contain an SAP username and password. See Defining Production Credentials.

2.1 Setting Up the Java Gateway

The Java Gateway server runs within a JVM, which can be on the same machine as InterSystems IRIS or on a different machine. Complete the following setup steps on the machine on which the Java Gateway will run:

1. Install the Java Runtime Environment (for example, JRE 1.8.0_67).
2. Make a note of the location of the installation directory for JRE. This is the directory that *contains* the subdirectories bin and lib.

   This is the value that you would use for *JAVA_HOME* environment variable. For example: `c:\Program Files\Java\jre8`

   You use this information later when you configure your production.
3. Also make a note of the Java version. If you are uncertain about the Java version, open a DOS window, go to the bin subdirectory of your Java installation, and enter the following command:

   ```
   java.exe -version
   ```

   You should receive output like the following, depending on your platform:

   ```
   java version "1.8.0_67"
   Java(TM) SE Runtime Environment (build 1.8.0_67-b24)
   Java HotSpot(TM) 64-Bit Server VM (build 23.19-b22, mixed mode)
   ```

It is not necessary to set any environment variables. To access the JVM, InterSystems IRIS uses information contained in the production.

2.2 Installing the SAP JCo Jar File

Obtain, from SAP, the SAP Java Connector 3.x, as appropriate for your operating system. Generally, this is provided as a compressed file. Uncompress it and place the contents in a convenient location. The directory should contain the following items:

- examples subdirectory
- javadoc subdirectory
- Readme.txt file
- sapjco3.dll file
- sapjco3.jar file
- sapjcomanifest.mf file

2.3 Generating Proxy Classes for SAP JCo

To communicate with SAP JCo, your interoperability-enabled namespace must contain proxy classes that represent SAP JCo. To generate these classes, do the following:

1. Start the Java Gateway.

   The easiest way to do this is as follows:

   a. Create a simple production that contains only one business host: EnsLib.JavaGateway.Service. See EnsLib.JavaGateway.Service Settings.

   b. Start the production, which starts the Java Gateway.

2. In the Terminal, change to your interoperability-enabled namespace and use the **ImportSAP()** method of EnsLib.SAP.BootStrap, as follows:

   ```
   do ##class(EnsLib.SAP.BootStrap).ImportSAP(pFullPathToSAPJarFile,pPort,pAddress)
   ```

   Where:

   - *pFullPathToSAPJarFile* is the full path to the SAP Jar file.
   - *pPort* is the port used by the Java Gateway.
   - *pAddress* is the IP address used by the Java Gateway.

2.4 Testing the SAP Connection

To test the SAP connection, do the following in the Terminal (or in code):

1. Create an instance of EnsLib.SAP.Utils, which is the class that the SAP business operation uses internally to connect to SAP.
2. Set the following properties of that instance. These are string properties unless otherwise noted.

- SAPClient—SAP Client e.g 000.
- SAPUser—Username that has access to the SAP server.
- SAPPassword—Password for the user.
- SAPLanguage
- SAPHost— Host name or IP address of the SAP server.
- SAPSystemNumber—SAP SystemNumber e.g 00.
- JavaGatewayAddress—IP address or name of the machine where the JVM to be used by the Java Gateway server is located. Or specify this as the name of the external language server for Java and leave JavaGatewayPort black. For external language servers, Managing External Server Connections.
- JavaGatewayPort—Port used by the Java Gateway. Or leave this blank if JavaGatewayAddress is an external language server name.
- SAPTransactionAutoCommit—Specifies whether to execute the BAPI "BAPI_TRANSACTION_COMMIT" after a successful BAPI/RFC-call. This property is %Boolean.

3. Call the PingSAP() method of your instance. This method connects to SAP and performs a dynamic invocation of the STFC_CONNECTION function. It returns a %Status.

3

Using the SAP Java Connector

This page describes how to add the required components to your production so that it can send requests to SAP. Also see Setup Tasks.

3.1 Basics

Add the following business hosts to your production.

- The business service EnsLib.JavaGateway.Service. Configure this business host as described later on this page.
- The business operation EnsLib.SAP.Operation.

  Configure this business host as described in later on this page.
- One or more business hosts that send SAP request messages to EnsLib.SAP.Operation, as needed.

  Use the message classes that you generated. Your business hosts should create instances of these classes, set properties as applicable, and send the messages to the instance of EnsLib.SAP.Operation.

3.2 Settings for EnsLib.JavaGateway.Service

Configure the settings for EnsLib.JavaGateway.Service so that it can find the Java Gateway. These settings are:

ExternalServerName

The value of this setting should be an external language server name as described in Managing External Server Connections.

Stop Named Gateway When Stopping

Determines if an attempt will be made to stop the specified named server when this business host stops. The default is off.

3.3 Settings for EnsLib.SAP.Operation

EnsLib.SAP.Operation sends requests to SAP JCo, via the Java Gateway. For this business host, specify the following settings:

SAPClient

SAP Client e.g 000.

SAPCredentials

This is the name of the set of production credentials to use when accessing the SAP server. See Defining Production Credentials.

SAPLanguage

SAPHost

Host name or IP address of the SAP server.

SAPSystemNumber

SAP SystemNumber e.g 00.

SAPTransactionAutoCommit

Specifies whether to execute the BAPI "BAPI_TRANSACTION_COMMIT" after a successful BAPI/RFC-call.

SAPResponseHandler

Configuration item in this production that should receive the SAP response.

JavaGatewayConfigItemName

Name of the (required) configuration item that hosts the Java Gateway.

For settings not listed here, see Settings in All Productions.